Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number : 11

Start date: 29 -Dec- 2025

Names:

Payal Shinde
Mohammad Saad
Jyothi Reddy
Gowtham Mani
Kumari Divya Sree

Internship Duration: 8 Weeks

1. Project Title

Visualizing Election Data for a News Company Using Power BI

2. Project Objective

The main objective of this internship project was to develop an interactive Election Data Visualization System using Power BI that enables news agencies to analyze election data efficiently. The project aimed to integrate election datasets from multiple sources and create dashboards that present meaningful insights such as vote share trends, voter turnout patterns, party performance, and constituency-level results.

In the fast-paced environment of election coverage, journalists require accurate and easily interpretable data visualizations to present real-time updates and historical analysis. This project helps editorial teams quickly understand election dynamics and communicate complex data clearly to viewers.

The solution focuses on data integration, modeling, and interactive visualization to support decision-making and reporting. By building scalable dashboards, the system can also be adapted for future elections and new datasets.

3. Project description in detail

The project involved developing an election analytics platform using Microsoft Power BI to visualize and analyze election data. Public datasets containing information about candidates, parties, constituencies, vote counts, election years, and voter turnout were collected in CSV and Excel formats and imported into Power BI using Power Query.

The data was cleaned by removing duplicates, handling missing values, and standardizing constituency and party names. A Star Schema data model was created with a Fact_ElectionResults

table and dimension tables such as Dim_Candidate, Dim_Constituency, Dim_Party, and Dim_Date to improve analysis and filtering.

Several interactive dashboards were developed, including Historical Trends, Real-Time Results, Turnout and Demographics, and Party & Candidate Performance dashboards. These dashboards visualize vote share trends, seat counts, voter participation patterns, and candidate performance using charts, heatmaps, and filters.

Overall, the project demonstrates how Power BI can transform complex election data into clear and interactive visual insights, helping news agencies analyze and present election results effectively.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project understanding and dataset collection	Collected election datasets and studied project requirements
Week 2	Data cleaning and modeling	Cleaned datasets and created star schema in Power BI
Week 3	Historical trends dashboard development	Built vote share and party trend visualizations
Week 4	Interactive filtering and analysis	Implemented slicers and dynamic filters
Week 5	Real-time results dashboard	Created simulated real-time election results visuals
Week 6	Constituency and party analysis	Added party-wise and candidate-wise analysis
Week 7	Turnout and demographic insights	Developed turnout and demographic dashboards
Week 8	Final dashboard integration and documentation	Completed final dashboards and prepared project report

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Understanding project scope and data sources	1 January 2026
Prototype / First Draft	Initial dashboard and data model created	27 January 2026
Mid-Term Review	Dashboard improvements and feedback implementation	10 February 2026
Final Submission	Completed dashboards and documentation	27 February 2026
Presentation	Final project presentation and demonstration	5 March 2026

5b. Project execution details

The project execution followed a structured development approach.

First, election datasets were gathered and analyzed to understand the available data attributes. The datasets were imported into Power BI using Power Query, where data cleaning and transformation were performed.

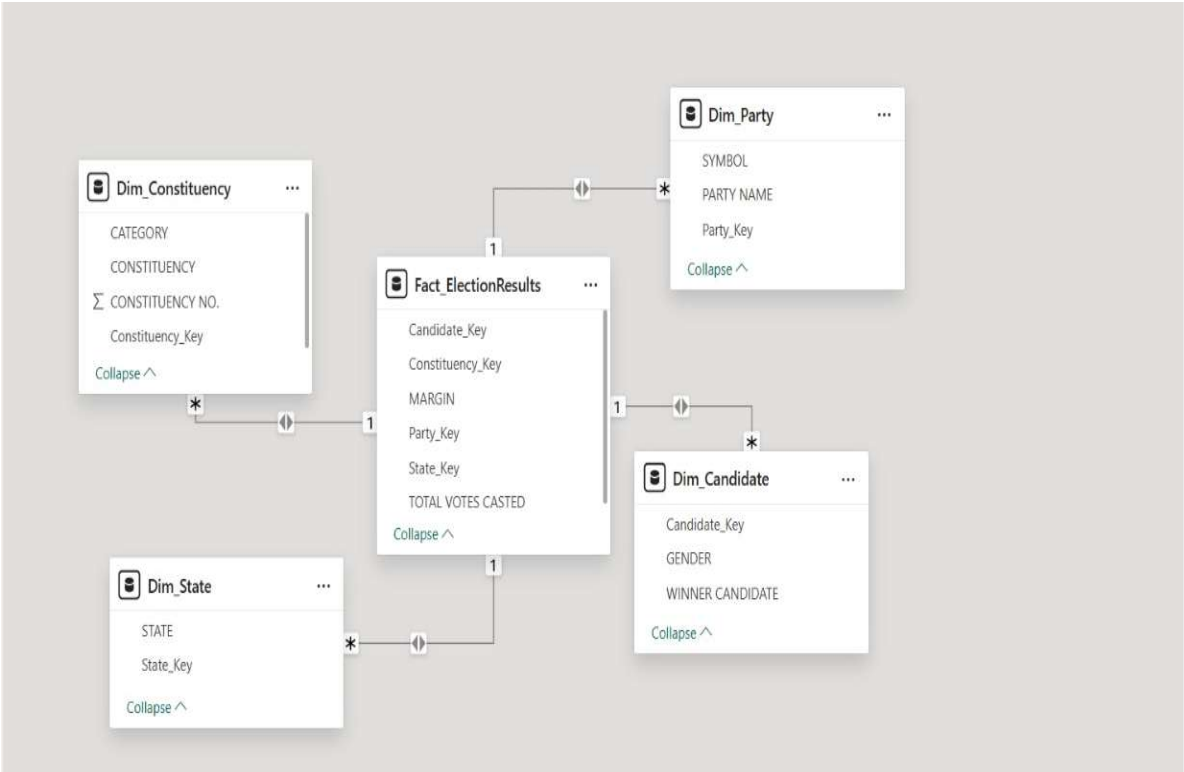
Next, a star schema data model was designed to organize the dataset efficiently. Relationships were established between fact and dimension tables to support accurate analysis.

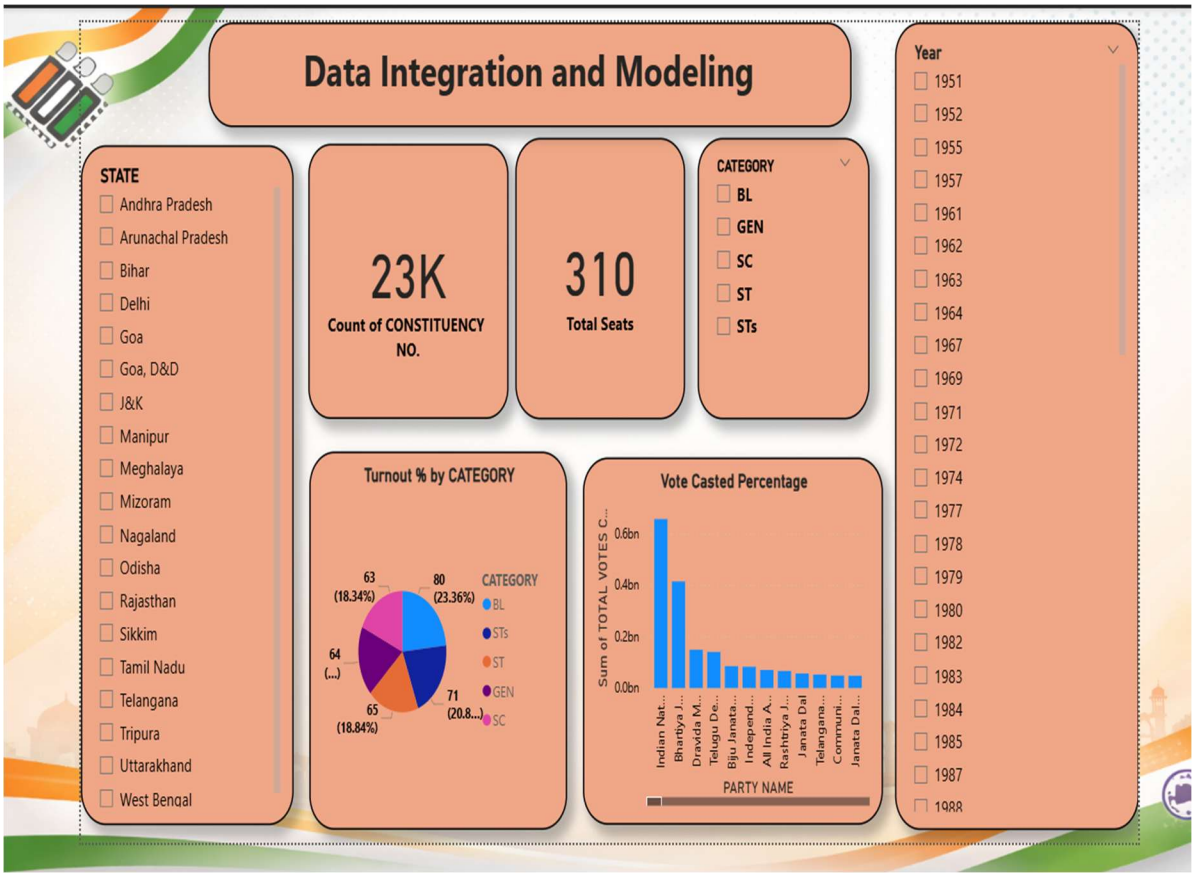
After building the data model, multiple dashboards were designed to address different analytical needs, such as historical trends, real-time results, and voter turnout insights.

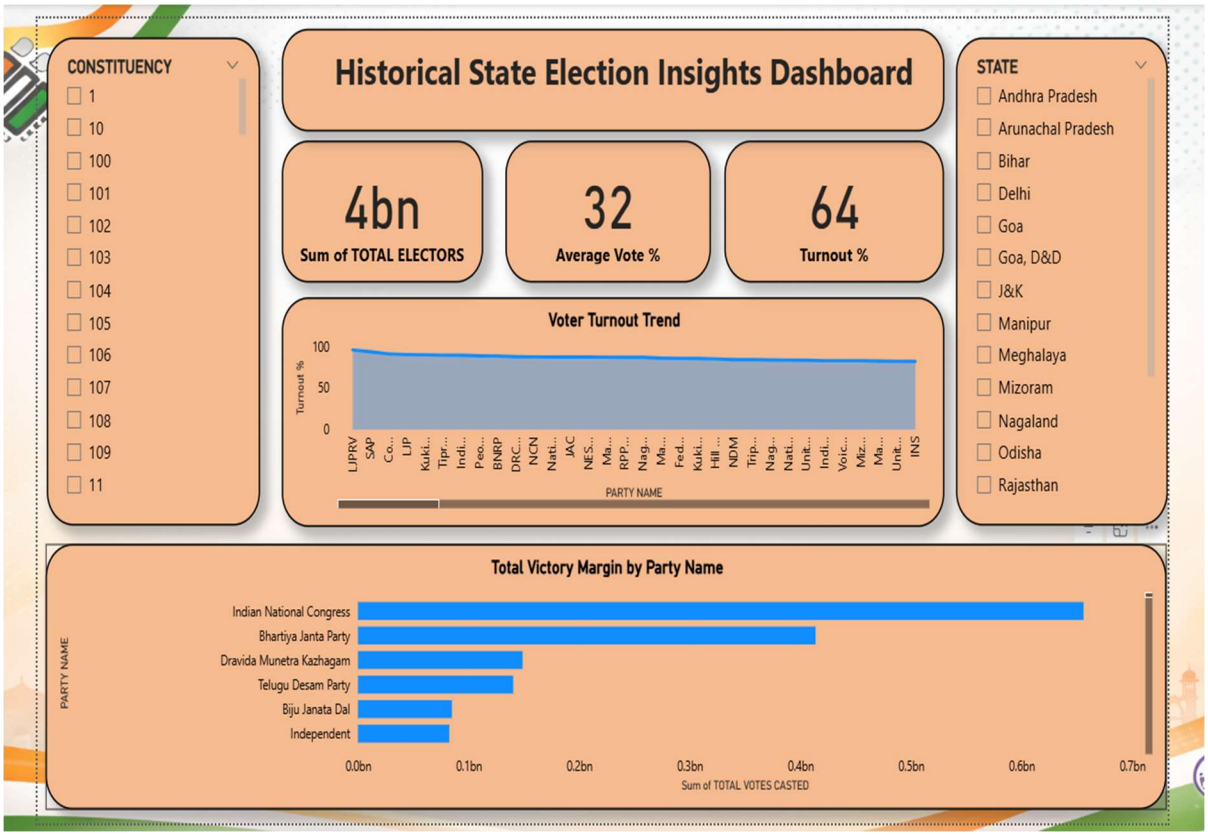
Each dashboard was developed using Power BI visualizations, including line charts, bar charts, heatmaps, and KPI indicators. Interactive slicers and filters were implemented to allow users to explore the data dynamically.

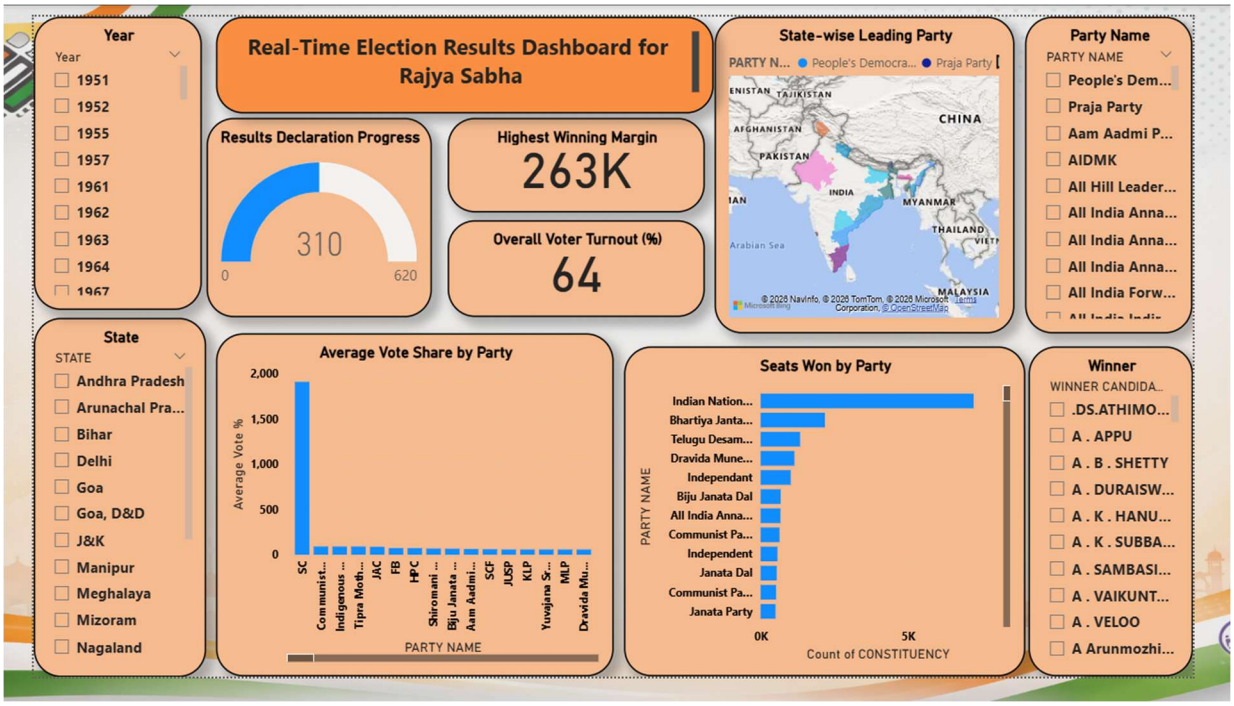
Finally, the dashboards were tested for accuracy and usability, and the results were documented in the final report.

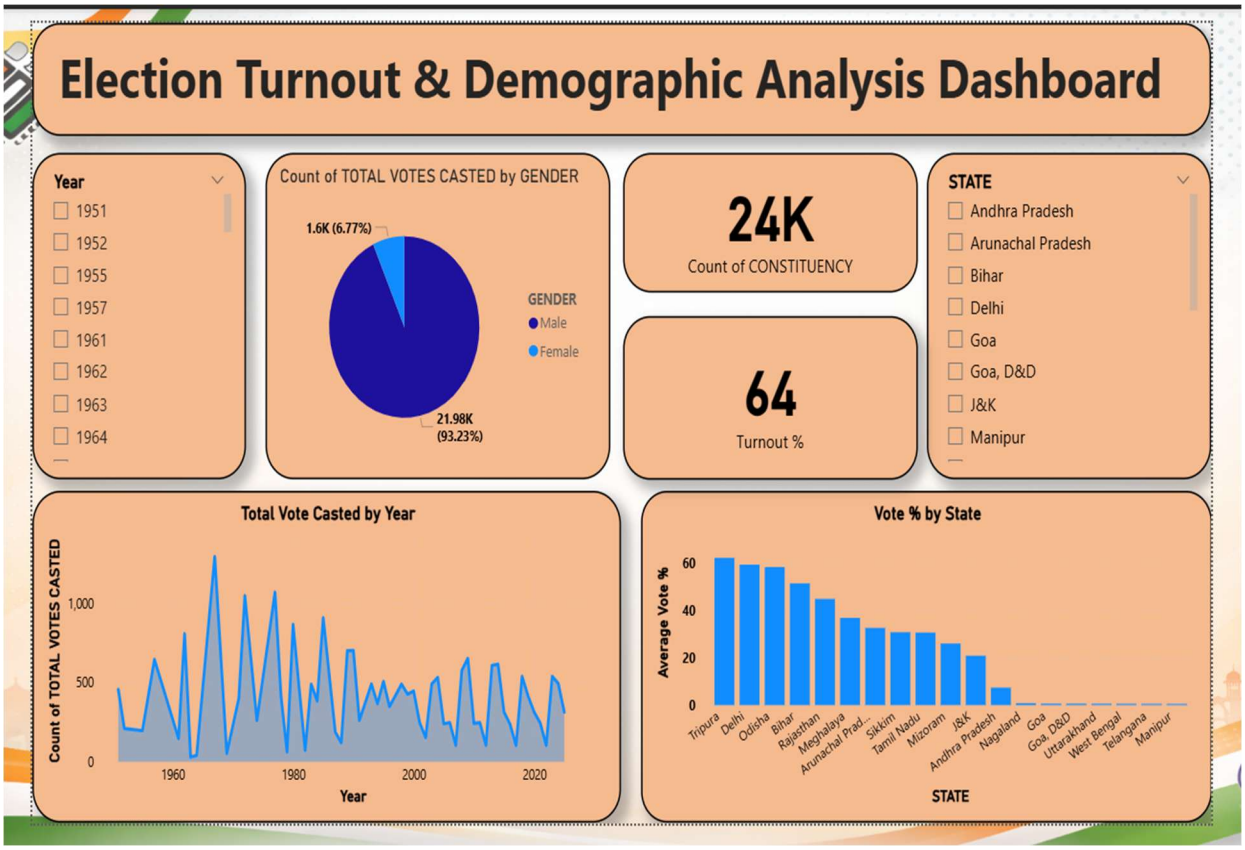
6. Snapshots / Screenshots

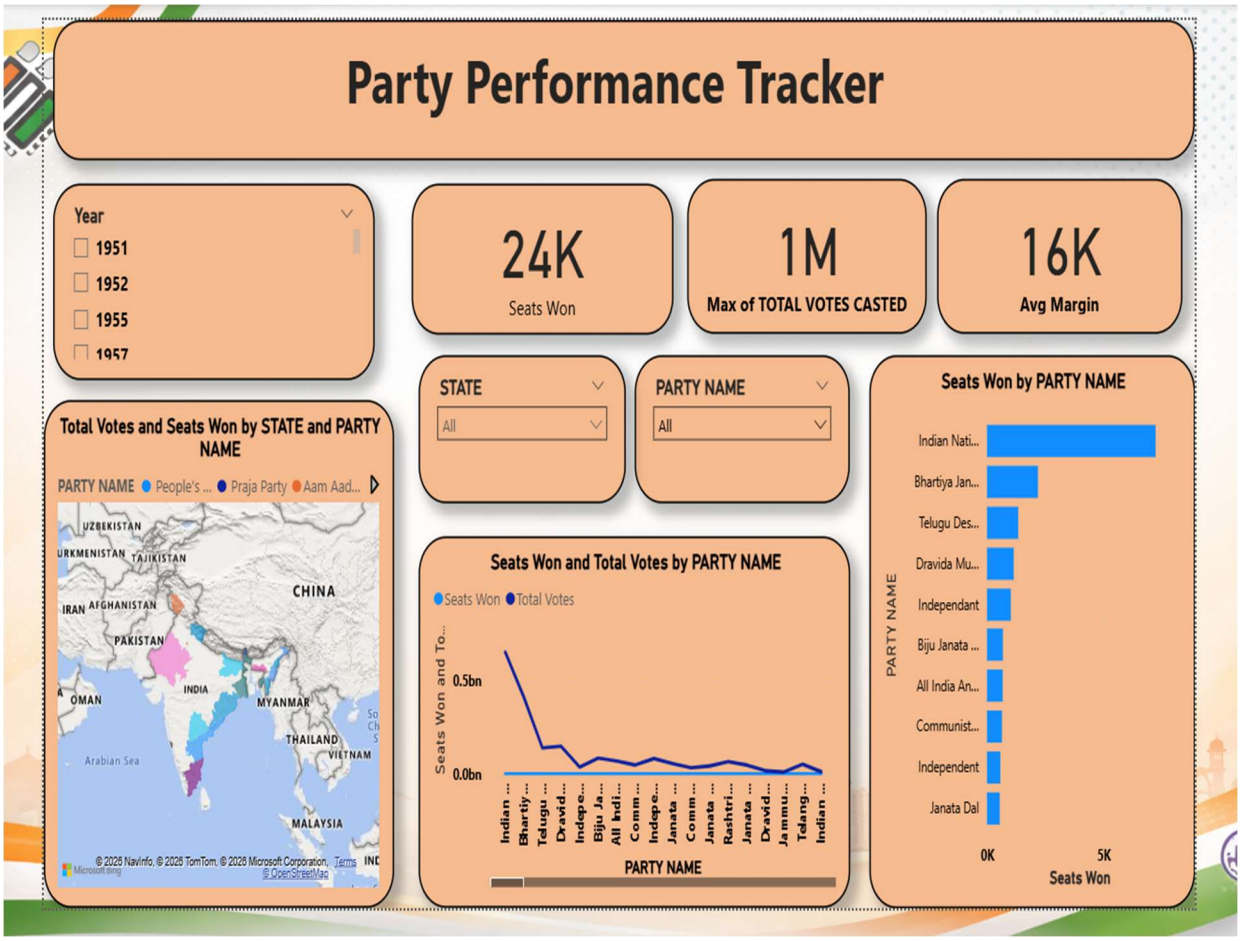



Data Integration and Modeling
STATE
Andhra Pradesh
Arunachal Pradesh
Bihar
Delhi
Goa
Goa, D&D
J&K
Manipur
Meghalaya
Mizoram
Nagaland
Odisha
Rajasthan
Sikkim
Tamil Nadu
Telangana
Tripura
Uttarakhand
West Bengal
23K
Count of CONSTITUENCY NO.
310
Total Seats
CATEGORY
BL
GEN
SC
ST
STs
Year
1951
1952
1955
1957
1961
1962
1963
1964
1967
1969
1971
1972
1974
1977
1978
1979
1980
1982
1983
1984
1985
1987
Turnout % by CATEGORY
63 (18.34%)
80 (23.36%)
64 (...)
65 (18.84%)
71 (20.8...)
CATEGORY
BL
STs
ST
GEN
SC
Vote Casted Percentage
Sum of TOTAL VOTES C...
0.6bn
0.4bn
0.2bn
0.0bn
Indian Nat...
Bhartiya J...
Dravida M...
Telugu De...
Biju Janata...
Independ...
All India A...
Rashtriya J...
Janata Dal
Telangana...
Communi...
Janata Dal...
PARTY NAME


CONSTITUENCY
1
10
100
101
102
103
104
105
106
107
108
109
11
Historical State Election Insights Dashboard
4bn
Sum of TOTAL ELECTORS
32
Average Vote %
64
Turnout %
Voter Turnout Trend
Turnout %
100
50
0
LJPRV
SAP
Co...
LJP
Kuki...
Tipr...
Indi...
Peo...
BNRP
DRC...
NCN
Nati...
JAC
NES...
Ma...
RPP...
Nag...
Ma...
Fed...
Kuki...
Hill ...
NDM
Trip...
Nag...
Nati...
Unit...
Indi...
Voic...
Miz...
Ma...
Unit...
INS
PARTY NAME
STATE
Andhra Pradesh
Arunachal Pradesh
Bihar
Delhi
Goa
Goa, D&D
J&K
Manipur
Meghalaya
Mizoram
Nagaland
Odisha
Rajasthan
Total Victory Margin by Party Name
PARTY NAME
Indian National Congress
Bhartiya Janta Party
Dravida Munetra Kazhagam
Telugu Desam Party
Biju Janata Dal
Independent
0.0bn
0.1bn
0.2bn
0.3bn
0.4bn
0.5bn
0.6bn
0.7bn
Sum of TOTAL VOTES CASTED


Real-Time Election Results Dashboard for Rajya Sabha
Year
Year
1951
1952
1955
1957
1961
1962
1963
1964
1967
Results Declaration Progress
310
0
620
Highest Winning Margin
263K
Overall Voter Turnout (%)
64
State-wise Leading Party
PARTY N... People's Democra... Praja Party
CHINA
AFGHANISTAN
PAKISTAN
INDIA
MYANMAR
THAILAND
Arabian Sea
MALAYSIA
Party Name
PARTY NAME
People's Dem...
Praja Party
Aam Aadmi P...
AIDMK
All Hill Leader...
All India Anna...
All India Anna...
All India Anna...
All India Forw...
State
STATE
Andhra Pradesh
Arunachal Pra...
Bihar
Delhi
Goa
Goa, D&D
J&K
Manipur
Meghalaya
Mizoram
Nagaland
Average Vote Share by Party
Average Vote %
2,000
1,500
1,000
500
0
SC
Communist...
Indigenous ...
Tipra Moth...
JAC
FB
HPC
Shiromani ...
Biju Janata ...
Aam Aadmi...
SCF
JUSP
KLP
Yuvajana Sr...
MLP
Dravida Mu...
PARTY NAME
Seats Won by Party
PARTY NAME
Indian Nation...
Bhartiya Janta...
Telugu Desam...
Dravida Mune...
Independant
Biju Janata Dal
All India Anna...
Communist Pa...
Independent
Janata Dal
Communist Pa...
Janata Party
0K
5K
Count of CONSTITUENCY
Winner
WINNER CANDIDA...
.DS.ATHIMO...
A . APPU
A . B . SHETTY
A . DURAISW...
A . K . HANU...
A . K . SUBBA...
A . SAMBASI...
A . VAIKUNT...
A . VELOO
A Arunmozhi...


Election Turnout & Demographic Analysis Dashboard
Year
1951
1952
1955
1957
1961
1962
1963
1964
Count of TOTAL VOTES CASTED by GENDER
1.6K (6.77%)
21.98K (93.23%)
GENDER
Male
Female
24K
Count of CONSTITUENCY
64
Turnout %
STATE
Andhra Pradesh
Arunachal Pradesh
Bihar
Delhi
Goa
Goa, D&D
J&K
Manipur
Total Vote Casted by Year
Count of TOTAL VOTES CASTED
1,000
500
0
1960
1980
2000
2020
Year
Vote % by State
Average Vote %
60
40
20
0
Tripura
Delhi
Odisha
Bihar
Rajasthan
Meghalaya
Arunachal Prad...
Sikkim
Tamil Nadu
Mizoram
J&K
Andhra Pradesh
Nagaland
Goa
Goa, D&D
Uttarakhand
West Bengal
Telangana
Manipur
STATE



7. Challenges Faced

Several challenges were encountered during the project:

Data Inconsistency

Election datasets contained inconsistent constituency names and missing values. These issues were resolved through data cleaning and standardization using Power Query.

Complex Data Relationships

Designing a proper star schema required careful planning to ensure accurate relationships between tables.

Visualization Design

Choosing the right visualizations to represent complex election data effectively required experimentation and optimization.

Performance Optimization

Large datasets sometimes affected dashboard performance, which was improved by optimizing data models and reducing unnecessary columns.

8. Learnings & Skills Acquired

This internship provided valuable technical and analytical experience.

Technical Skills

- Power BI dashboard development
- Data modeling using star schema
- Data transformation using Power Query
- Interactive data visualization techniques

Analytical Skills

- Election data analysis
- Trend identification
- Data storytelling

Soft Skills

- Team collaboration
- Problem solving
- Presentation and communication

9. Testimonials from team

Working on this project was a valuable and enriching experience. The project allowed us to apply theoretical knowledge to a real-world problem and develop practical data visualization skills.

The collaboration among team members helped improve problem-solving and communication skills. Overall, it was a great opportunity to learn modern data analytics tools and understand how data can support journalism and public information.

10. Conclusion

The internship project successfully demonstrated the use of Power BI for election data analysis and visualization. The developed dashboards provide meaningful insights into vote shares, turnout patterns, and party performance.

The project highlights the importance of data-driven journalism and how interactive visualizations can enhance election reporting. This experience also strengthened technical skills in data modeling, visualization, and analytics.

Overall, the internship was a highly valuable learning experience that contributes to future career opportunities in data analytics and information technology.

11. Acknowledgements

I would like to express my sincere gratitude to Infosys Springboard for providing this valuable internship opportunity. I also thank the mentors and instructors for their guidance throughout the program.

I am grateful to my teammates for their collaboration and support during the project. Their contributions and teamwork played a significant role in the successful completion of this internship project.

12. Github Repository :

https://github.com/Payal1726/Infosys_ElectViz-Election-Data-Visualization-for-Media-Team-A-